

Credit risk prediction system

Machine learning 622: 402418781

Current Problem

In recent years, financial institutions have increasingly relied on data-driven decision-making to manage credit risk more effectively, which often depend on fixed scoring rules and manual evaluation, These financial institutions are limited in their ability to adapt to complex and evolving financial behaviours. Where systems are incapable of obtaining complex relationships between variables, can result in false loan approvals and rejections as well as they are unable to adapt to fluctuating customer behaviour





Possible Solution

these financial institutions desperately need to make accurate loan approval decisions while supervising credit risk. A possible solution to these problems could be implementing a supervised machine learning classification system which be used to classify loan applicants according to a customers status.

In this presentation I aim to :

- Improve prediction accuracy
- Use PCA to reduce the number of features
- Compare the different ML models

Ensuring that financial institutions systems needs are met and work efficiently for each company regardless of fluctuating behaviour.

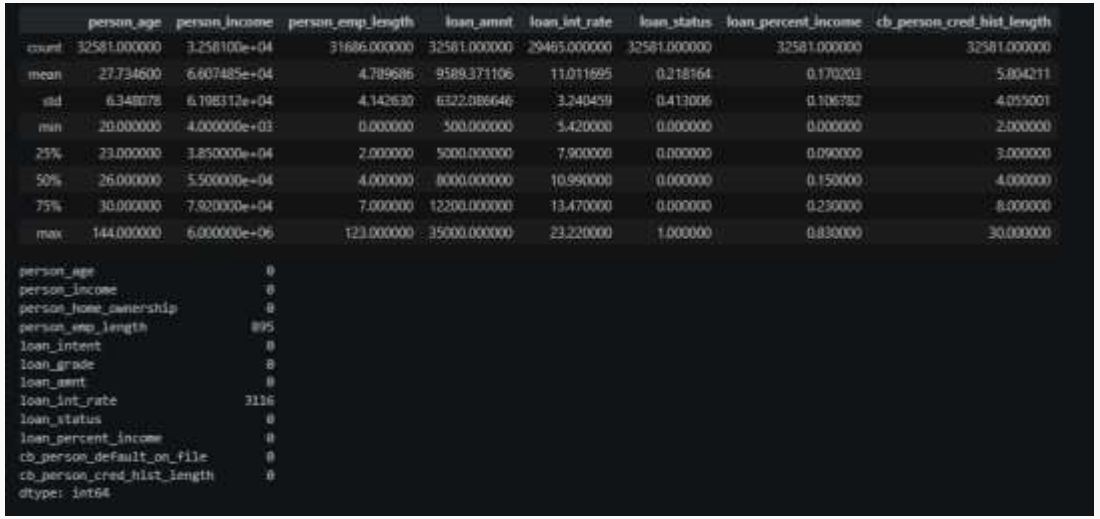
Dataset Overview

This project uses a loan application dataset with information about the applicants looking for a loan and their outcomes. This dataset also included :

- Applicant information (age, income, home ownership)
- Financial information
- Loan information
- Credit-related information
- And target variable

This dataset contained numerical and categorical variables with missing values which needed to be handled. This dataset was also checked for duplicate records which could cause a bias outcome and outlier. Categorical variables required encoding to numerical values, the scale of numerical variables was not uniform, hence feature scaling was done. Later PCA was applied to reduce the dimensionality of the feature. The objective is to use the available applicant(client) information to predict the loan_status of a loan application

DATASET DESCRIPTION



	person_age	person_income	person_emp_length	loan_amnt	loan_int_rate	loan_status	loan_percent_income	cb_person_cred_hist_length
count	32581.000000	3.258100e+04	31686.000000	32581.000000	29465.000000	32581.000000	32581.000000	32581.000000
mean	27.734600	6.607485e+04	4.789686	9589.371106	11.011695	0.218164	0.170203	5.804211
std	6.348078	6.198312e+04	4.142630	6322.086646	3.240459	0.413006	0.106782	4.055001
min	20.000000	4.000000e+03	0.000000	500.000000	5.420000	0.000000	0.000000	2.000000
25%	23.000000	3.850000e+04	2.000000	5000.000000	7.900000	0.000000	0.090000	3.000000
50%	26.000000	5.500000e+04	4.000000	8000.000000	10.990000	0.000000	0.150000	4.000000
75%	30.000000	7.920000e+04	7.000000	12200.000000	13.470000	0.000000	0.230000	8.000000
max	144.000000	6.000000e+06	123.000000	35000.000000	23.220000	1.000000	0.830000	30.000000

```
person_age                    0
person_income                 0
person_home_ownership         0
person_emp_length           895
loan_intent                   0
loan_grade                    0
loan_amnt                     0
loan_int_rate              3116
loan_status                   0
loan_percent_income           0
cb_person_default_on_file     0
cb_person_cred_hist_length    0
dtype: int64
```

Data Issues Identified

During the analysis phase(EDA), a number of issues regarding data quality were considered. The issues found were :

- missing values in some variables
- duplicate data records were examined and removed
- there was present outliers in numerical variables
- Categorical variables are not suitable for ML algorithms
- There were different scales and ranges in numerical variables

These issues if not handled can impact the performance of machine learning model and prediction results if ignored, such as a bias, inaccurate result false approval or rejection or a true rejection which can cause confusion, less productivity within the financial institution.



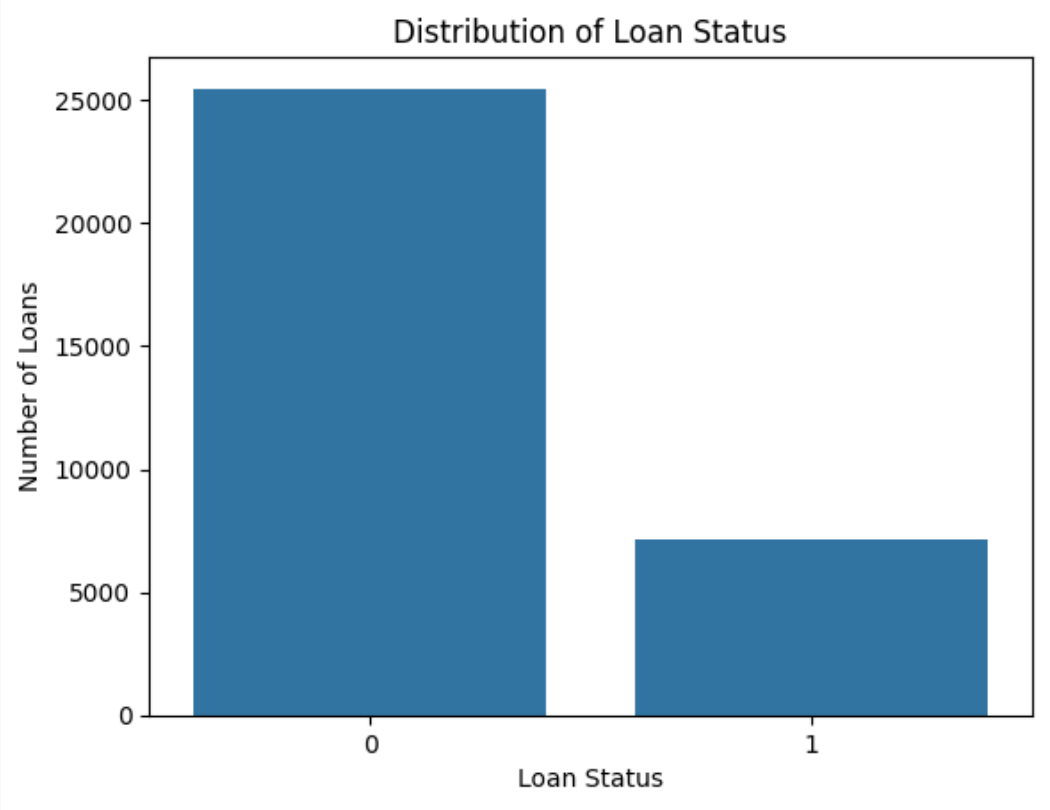
Preprocessing summary

Before the dataset could be analysed for customer outcomes and a generalisation, the dataset was prepared before developing the machine learning models through a 6-step process I call FROCFT:

- **F**ind the missing values: This is where missing values were identified and handled to ensure the algorithm could process the data properly as I noticed when missing values are still visible within the dataset it affects the outcome as well as the PCA .
- **R**emove duplicate records: Duplicates which in the model and can lead to overfitting. Removing duplicates can make sure that the model is trained on unique data values which can improve its performance.
- **O**utliers: The IQR method was used to identify extreme numerical values; the outliers were treated using appropriate limits.
- **C**ategorical Encoding: Categorical variables were converted to numerical values using encoding as categorical features are not suitable for complex machine learning algorithms.
- **F**eature Scaling: This is where numerical features were standardized using StandardScalar. Feature scaling is important because numerical features can have different ranges and units. StandardScaler puts the numerical features on a similar scale this makes sure features with large values from having too much influence on a machine learning model.
- **T**rain-Test Split: This is where the Pareto Principle is applied, the improved cleaned dataset was then divided into 80% training data and 20% testing to ensure that the financial institution system learns from the most significant factors

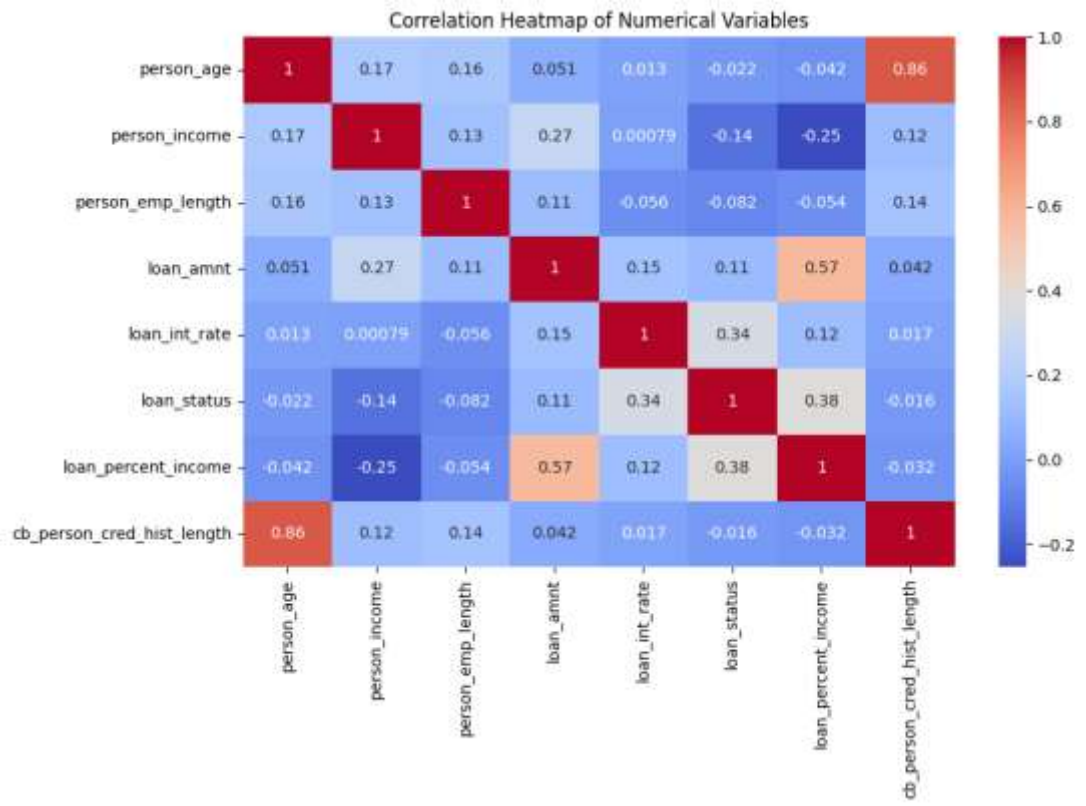
Exploratory Data Analysis(EDA)

TARGET VARIABLE DISTRIBUTION



This visual representation about the loan_status shows the distribution of the 2 loan-status classes (0:approved and 1:rejected) where 25000+ records are approved and 70000+records where denied. There is an imbalance due to more observations for class 0 than class 1. This imbalance should be considered when evaluating ML models, this accuracy alone may not give complete picture of model performance

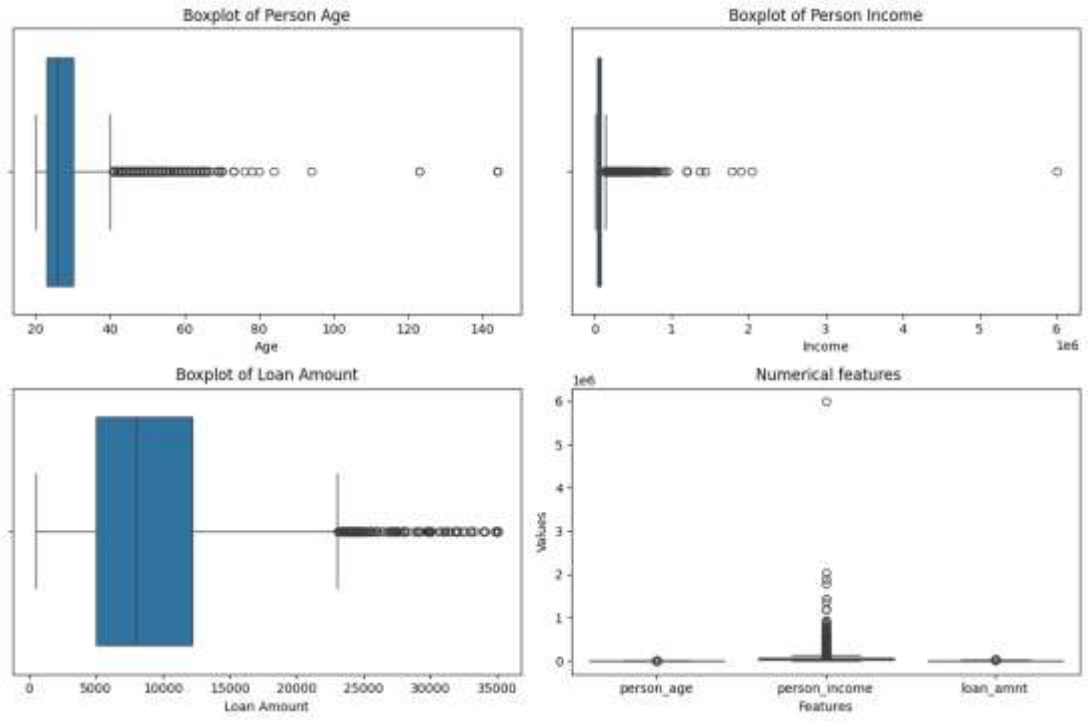
CORRELATION HEATMAP



This representation indicates the correlation between the numerical variables within the data. As one can see person_age and cb_person_cred_hist_length have a strong positive correlation (0.86). Loan_amnt and loan_percent_income have a positive correlation (0.57). Loan_percent_income with 0.38. Loan_status is correlated to loan_int_rate with the value 0.34.This heatmap shows that some numerical variables have stronger relationships than others. The strongest relationship is between person_age and crdt_hist_length, however this is not an indication that one variable causes another

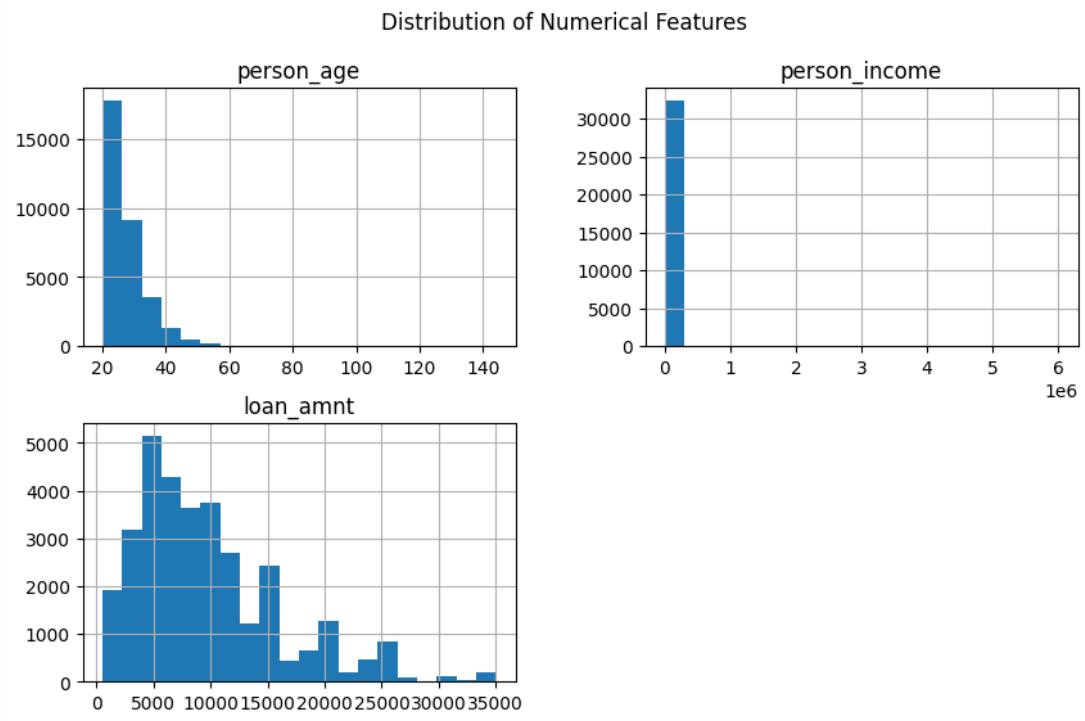
Exploratory Data Analysis (EDA)

BOXPLOTS FOR OUTLIER



This visual representation identifies the potential outliers in age, income and loan amount, the graphs show that person income has a large number of extreme values above the upper whisker while, person age and loan amount have values outside their range

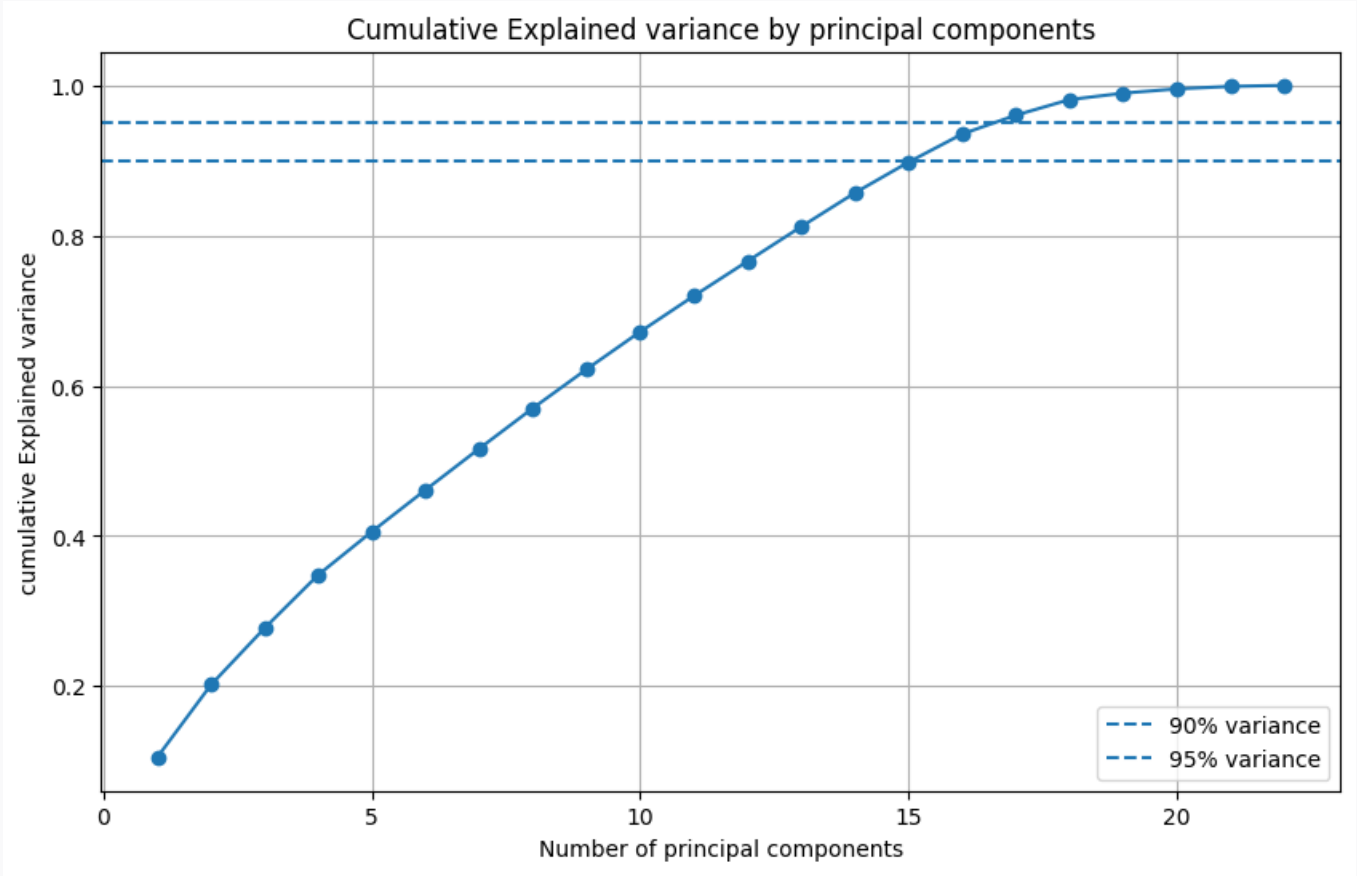
HISTOGRAM OF NUMERICAL FEATURES



These 3 histograms display the distribution of frequencies of numeric variables in the dataset. We can see from the histograms where most of the observations lie and determine the distribution shapes of all the variables. Person_age is right-skewed where most clients age are mostly younger than 60. person_income is strongly right-skewed where at lower income values a long tail is extending towards high incomes meaning clients who applied for a loan don't earn that much and loan_amnt is right-skewed at lower-to-middle values and fewer observations at much higher loan amounts

Principal component Analysis

Geeksforgeeks defines PCA as "***a dimensionality reduction technique and helps us to reduce the number of features in a dataset while keeping the most important information***". Within the assignment PCA transformed the original features into a smaller number of principal components while maintaining as much variance as possible. This graphical representation has 22 features after preprocessing, PCA shows that you don't need all 22 features to keep most of the information as PCA reduces dimensionality from 22 components to 17 components while retaining 95% of the information as using only 17 features can retain 95% of the variance compared to keeping only 16 features which maintains 90%. Therefore as the number of features increases more information from the original dataset is retained



This is a graphical representation of PCA and variance plot

Model before PCA

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KNN

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Accuracy: 0.8918
Precision: 0.8888
Recall: 0.8918
f1-score: 0.8848
Confussion Matrix:
[[4332  131]
 [ 486  752]]
Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.97      0.93      4463
           1       0.85      0.61      0.71      1238

    accuracy                           0.89      5701
   macro avg       0.88      0.79      0.82      5701
weighted avg       0.89      0.89      0.88      5701


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Decision tree

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Accuracy: 0.8832
Precision: 0.886
Recall: 0.8832
f1-score: 0.8844
```

```
==========Logistic Regression==================

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Logistic Regression

==========================
Accuracy: 0.8613
Precision: 0.8533
Recall: 0.8613
f1-score: 0.8531
Confussion Matrix:
[[4231  232]
 [ 559  679]]
Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.95      0.91      4463
           1       0.75      0.55      0.63      1238

    accuracy                           0.86      5701
   macro avg       0.81      0.75      0.77      5701
weighted avg       0.85      0.86      0.85      5701


========================
KNN

==========================
Accuracy: 0.8918
```

3 models before PCA was applied were trained using the original scaled features. Knn has the best results with 89.18% and decision tree 88.32% and logistic regression with 86.13%

Model after PCA

```
======================Model Results before PCA ==========================================
Accuracy: 0.8918
Precision: 0.8888
Recall: 0.8918
F1-score: 0.8848
Decision Tree

======================Model Results before PCA ==========================================
Accuracy: 0.8832
Precision: 0.886
Recall: 0.8832
F1-score: 0.8844
Confussion Matrix:
[[4093  370]
 [ 296  942]]
Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.92      0.92      4463
           1       0.72      0.76      0.74      1238

    accuracy                           0.88      5701
   macro avg       0.83      0.84      0.83      5701
weighted avg       0.89      0.88      0.88      5701
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.95      0.91      4463
           1       0.75      0.52      0.61      1238

    accuracy                           0.86      5701
   macro avg       0.82      0.74      0.76      5701
weighted avg       0.85      0.86      0.85      5701
```

Final Model Comparison

```
======================Final Model Comparison==========================
                   Model  Accuracy  Precision  Recall  f1_Score
0    Logistic Regression    0.8613     0.8533  0.8613    0.8531
1                    KNN    0.8918     0.8888  0.8918    0.8848
2          Decision tree    0.8832     0.8860  0.8832    0.8844
3  Logistic Regression PCA    0.8588     0.8505  0.8588    0.8487
4                KNN PCA    0.8925     0.8891  0.8925    0.8862
5       Decision tree PCA    0.8337     0.8347  0.8337    0.8342
```

After the application of PCA, the model that had the highest accuracy was knn algorithm with 89.25% then decision tree with 88.logistic regression with 85.88% then decision tee with 83.37% . All models worked effectively with the original scaled features, this could mean that
Knn and logistic regression managed the feature patterns better based o accuracy and f1_scores

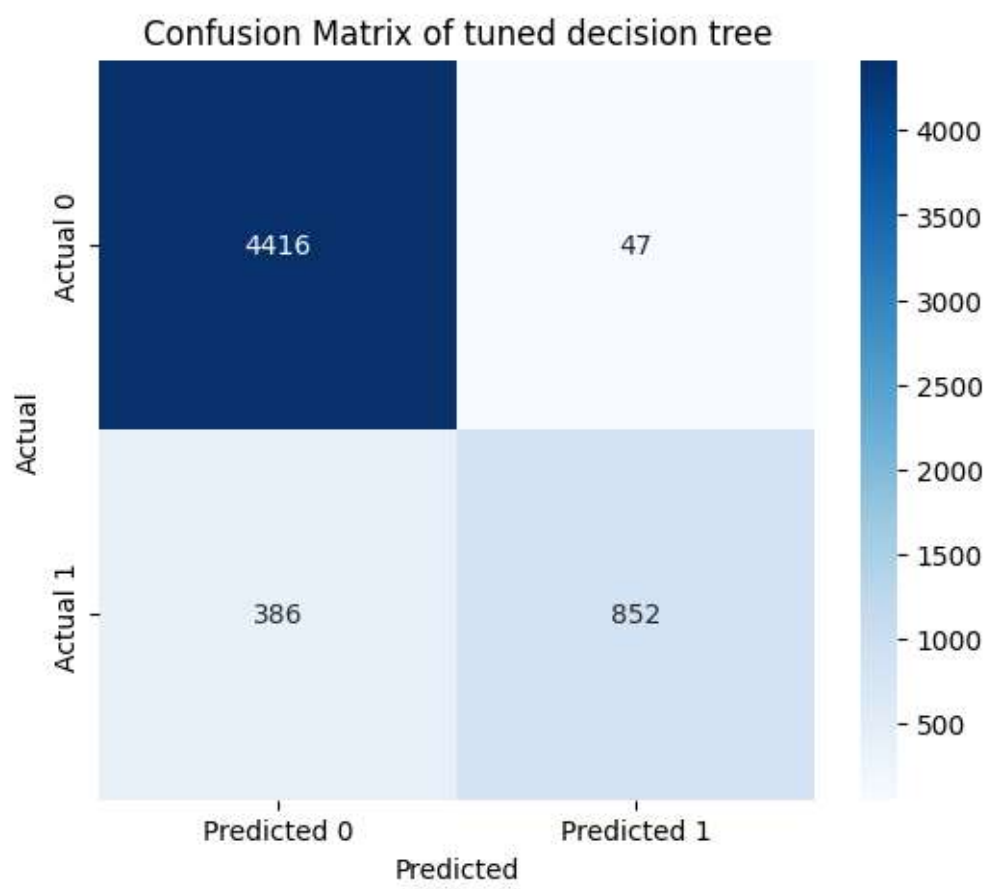
Hyperparameter tuning

==========Before vs After Tuning==========

	Metric	Before Tuning	After Tuning
0	Accuracy	0.8832	0.9240
1	Precision	0.8860	0.9257
2	Recall	0.8832	0.9240
3	f1-Score	0.8844	0.9194

- GridSearchCV was used to find the best combination of hyperparameters, improve the models performance as well as reduce any overfitting possibilities. To tune the model, I used the decision tree algorithm and performed GridSearchCV to try out different combinations of hyperparameters. The purpose was to identify different combinations of hyperparameters would yield both good prediction performance and control over the complexity. Some of the parameters have maximum depth of tree , number of samples needed to split and the minimum number of samples per leaf and criterion for splitting. After that step I compared the tuned and initial decision tree

Visual representation of a tuned confusion matrix



This representation demonstrates how many predictions were right and wrong. The left-top box and right-bottom box shows the correct predictions (TP, TN) where the right-top and right-bottom are predictions (FP,FN) which are incorrect classifications which can be very dangerous to the financial institution systems when approving or rejecting clients

IN CONCLUSSION….

- The credit Risk Prediction System effectively used the whole ML process from data preprocessing and exploratory data analysis to feature scaling PCA, model building and evaluating . 3 classification models were tested before and after using PCA. The credit risk prediction system proves that ML can be applied to classification of clients based on their financial records. It was important that more than one metric to evaluate model performance was used, since incorrect credit-risk classifications have different consequences. Therefore all 3 models were efficient yet all had different reactions to PCA where, knn had the most positive reaction to PCA compared to logistic regression and decision tree. The confusion matrix helped me understand correct and incorrect classification by each model. The model was optimised with the help of hyperparameter tuning.
- Therefore this new credit prediction system to accurately accept or reject applicants loans as well as accommodate fluctuating behaviour solves financial institutions problems while improving prediction accuracy.



Bibliography

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- Pareto Principal: Pareto Principle (The 80-20 Rule)

THANK YOU FOR READING!



As tony starks once said ***”I am Iron man… but even iron man needs a good model***”

…Maybe it sounds like something he said